4.3 Space with Functions

Definition 4.3.1 (Space with functions)
*Let X be a topological space and $\mathcal{O}_X$ be a subsheaf of the $\overline{k}$-valued functions on X (which is a sheaf of k-algebras). Then we say $(X, \mathcal{O}_X)$ is a **space with functions**. Additionally it's **local** if it satisfies the condition*

a) For every $f \in \mathcal{O}_X(U)$ we have $D(f) := \{f \neq 0\}$ is open and $\frac{1}{f} \in \mathcal{O}_X(D(f))$.

Note that the sheaf restriction maps $\rho_{UV} : \mathcal{O}_X(U) \to \mathcal{O}_X(V)$ correspond simply to restrictions of functions.

In the latter case X is a locally ringed space over k because $\mathcal{O}_{X,x}$ is a local k-algebra with maximal ideal $\mathfrak{m}_{X,x}$ consisting of stalks vanishing at x.

*A **regular morphism** or **regular map** $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a continuous function $\phi : X \to Y$ such that*

$$g \in \mathcal{O}_Y(U) \implies g \circ \phi \in \mathcal{O}_X(\phi^{-1}(U))$$

This determines a morphism of sheaves

$$\phi^\sharp : \mathcal{O}_Y \longrightarrow \phi_\star \mathcal{O}_X$$

*We say a regular morphism is **dominant** if $\phi(X)$ is dense in Y.*

An isomorphism is a homeomorphism and induces an isomorphism of sheaves, or equivalently has a two-sided regular inverse.

Proposition 4.3.2
The composition of regular maps is well-defined and associative.

4.3.1 Local Rings

Definition 4.3.3 (Local Ring)
*Let X be a local space with functions and $W \subset X$ an irreducible closed subset. Then we define the **local ring** at W to be (see (4.2.5))*

$$\mathcal{O}_{X,W} := \varinjlim_{U \cap W \neq \emptyset} \mathcal{O}_X(U)$$

If $\overline{W} = \overline{W'}$ then $\mathcal{O}_{X,W} = \mathcal{O}_{X,W'}$, so for example $\mathcal{O}_{X,W} = \mathcal{O}_{X,\overline{W}}$.

It is a local ring with unique maximal ideal

$$\mathfrak{m}_{X,W} := \{[(V, \sigma)] \mid V \cap W \neq \emptyset, \sigma \in \mathcal{O}_X(V), \sigma(x) = 0 \; \forall x \in V \cap W\}$$

*Define the **residue field***

$$k(\mathfrak{m}_{X,W}) := \mathcal{O}_{X,W}/\mathfrak{m}_{X,W}$$

In the case $W = \{x\}$ we write $\mathcal{O}_{X,x}$, $\mathfrak{m}_{X,x}$ and $k(\mathfrak{m}_{X,x})$.

Proof. Consider $(V, \sigma) \in \mathcal{O}_{X,W} \setminus \mathfrak{m}_{X,W}$. Then by definition $D(\sigma) \cap W \neq \emptyset$ and $(D(\sigma), \sigma^{-1})$ is an explicit inverse. ∎

Corollary 4.3.4 (Field of Rational Functions)
*Let X be an irreducible space with functions. Then $\mathcal{O}_{X,X} =: k(X)$ is a field, which we call the **field of rational functions**.*

Proposition 4.3.5 (Stalk Maps)
Suppose $\phi : X \to Y$ is a regular morphism and $Z \subset \overline{\phi(W)}$. Then there exists a unique (local) k-algebra homomorphism $\phi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}$ such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\rho \circ \phi_V^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,Z} & \dashrightarrow & \mathcal{O}_{X,W} \end{array}$$

for all open sets $V \subset Y$ for which $V \cap Z \neq \emptyset$ and some $U \subset \phi^{-1}(V)$ for which $U \cap W \neq \emptyset$. This is functorial in the sense that

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$$(\psi \circ \phi)_W = \psi_Z \circ \phi_W$$

Proof. If V meets Z then by (4.1.19) $\phi^{-1}(V)$ meets W. Therefore the vertical maps are well-defined. We define

$$\phi_W((V', \sigma)) := (\phi^{-1}(V'), \sigma \circ \phi)$$

and immediately verify it is well-defined and ensures the diagram commutes. ∎

Proposition 4.3.6 (Stalk Maps II)
Let $\phi : X \to Y$ be a regular morphism of local spaces with functions. Suppose $y \in \overline{\{\phi(x)\}}$. Then there is a unique local algebra homomorphism $\phi_x : \mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_{X,x}$ making the following diagram commute

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi^\sharp_V} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,\phi(x)} & \dashrightarrow & \mathcal{O}_{X,x} \end{array}$$

for all open neighbourhoods V of $\phi(x)$ and some open neighbourhood $U \subset \phi^{-1}(V)$ of x. This is functorial in the sense that

$$(\psi \circ \phi)_x = \psi_{\phi(x)} \circ \phi_x$$

Proof. This follows immediately from (4.3.5). ∎

Proposition 4.3.7 (Stalk Maps Function Field)
Let X, Y be irreducible local spaces with functions and $\phi : X \to Y$ a dominant regular morphism. Then there is a unique homomorphism $\phi_\star : k(Y) \hookrightarrow k(X)$ making the following diagram commute

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi^\sharp_V} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ k(Y) & \overset{\phi_\star}{\hookrightarrow\!\!\dashrightarrow} & k(X) \end{array}$$

for all non-empty open subsets $V \subset Y$ and some non-empty open subset $U \subset \phi^{-1}(V)$. This is functorial in the sense that

$$(\phi \circ \psi)_\star = \psi_\star \circ \phi_\star$$

Proof. This follows immediately from (4.3.5) with $W = X$ and $T = Y$. ∎

Proposition 4.3.8
Let X be a local space with functions and Z, W be irreducible subsets such that $Z \subset \overline{W}$. Then there is a canonical homomorphism $\mathcal{O}_{X,Z} \to \mathcal{O}_{X,W}$ making the following diagram commute

$$\begin{array}{ccc} \mathcal{O}_X(U) & & \\ \downarrow & \searrow & \\ \mathcal{O}_{X,Z} & \dashrightarrow & \mathcal{O}_{X,W} \end{array}$$

for all open subsets U of X meeting Z.

This holds in particular when X is irreducible we may take $W = X$ and $\mathcal{O}_{X,W} = k(X)$.

Proof. This follows immediately from (4.3.5) by considering the case $\phi = 1_X$. ∎

Proposition 4.3.9 (Stalk Maps under Specialisation)
Let X, Y be irreducible local spaces with functions, $\phi : X \to Y$ a dominant regular morphism and $x \in X$ such that $y \in \overline{\{\phi(x)\}}$. Then for all neighbourhoods V of $\phi(x)$ we have a commutative diagram

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$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(\phi^{-1}(V)) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,y} & \xrightarrow{\phi_x} & \mathcal{O}_{X,x} \\ \downarrow & & \downarrow \\ k(Y) & \xrightarrow{\phi_\star} & k(X) \end{array}$$

Lemma 4.3.10 (Evaluation of Stalks)
Let X be a local space of functions and $x \in X$. Then there is a unique function

$$\mathrm{ev}_x : k\big(\mathfrak{m}_{X,x}\big) \hookrightarrow \overline{k}$$

making the following diagram commute for all open neighbourhoods U of x

$$\begin{array}{ccccc} \mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_{X,x} & \longrightarrow & k\big(\mathfrak{m}_{X,x}\big) \\ & \searrow^{\mathrm{ev}_x} & & & \downarrow{\scriptstyle \mathrm{ev}_x} \\ & & & & \overline{k} \end{array}$$

This is functorial in the sense that for a regular morphism $\phi : X \to Y$ there is a commutative diagram

$$\begin{array}{ccc} k\big(\mathfrak{m}_{Y,\phi(x)}\big) & \xrightarrow{\phi_x} & k\big(\mathfrak{m}_{X,x}\big) \\ \downarrow{\scriptstyle \mathrm{ev}_{\phi(x)}} & & \downarrow{\scriptstyle \mathrm{ev}_x} \\ \overline{k} & = & \overline{k} \end{array}$$

4.3.2 Open Immersions

Definition 4.3.11 (Open Immersion)
*Let X, Y be spaces with functions. Then we say a regular map $\phi : X \to Y$ is an **open immersion** if*

a) ϕ is an open embedding, and
b) ϕ induces an isomorphism of sheaves

$$\phi_\star\big(\mathcal{O}_Y|_{\phi(X)}\big) \xrightarrow{\sim} \mathcal{O}_X$$

Proposition 4.3.12 (Open Subspace)
*Let X be a (local) space of functions and $U \subset X$ an open subset. Then $(U, \mathcal{O}_X|_U)$ is a (local) space of functions which we call an **open subspace**. We may write $\mathcal{O}_U \coloneqq \mathcal{O}_X|_U$.*

The inclusion $i : U \hookrightarrow X$ is an open immersion and an open immersion $\phi : X \to Y$ induces an isomorphism between X and the open subvariety $\phi(X)$.

Proposition 4.3.13
Let X be a space of functions and $V \subset U$ open subsets. Then $(V, \mathcal{O}_U|_V) = (V, \mathcal{O}_X|_V)$.

Proposition 4.3.14
Let $\phi : X \to Y$ be an open embedding and $Z \subset Y$ closed. Then Z is irreducible iff $\phi^{-1}(Z)$ is irreducible.

In particular if Y is irreducible then so is X.

Proof. We may reduce to the case $X = U$ is an open subset of Y. Then it follows from (4.1.73) and (4.1.70). ∎

Proposition 4.3.15
Let $\phi : X \to Y$ be a regular open embedding. Then the following are equivalent

a) ϕ is an open immersion
b) for every irreducible subset $W \subset X$ the canonical map

$$\mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}$$

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is an isomorphism for all Z such that $\overline{Z} = \overline{\phi(W)}$

c) b) holds for some Z such that $\overline{Z} = \overline{\phi(W)}$

In particular for all open subsets $V \subset Y$ and $U \subset \phi^{-1}(V)$ such that $U \cap W \neq \emptyset$ we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,\phi(W)} & \xrightarrow{\sim} & \mathcal{O}_{X,W} \end{array}$$

Proof. *b)* $\Longrightarrow$ *a)* By definition we may factor into a regular homeomorphism $X \xrightarrow{\sim} V \subset Y$. It is an isomorphism of spaces by (4.2.11).

a) $\Longrightarrow$ *b)* We may reduce to the case that $X \subset Y$ is an open subset. Because an open subset of X intersects W if and only if it intersects Z we find that the map is bijective.

c) $\Longleftrightarrow$ *b)* we simply observe that if $\overline{Z} = \overline{Z'}$ then the canonical maps are the same. ∎

Proposition 4.3.16
Let $\phi : X \to Y$ be a dominant open immersion. If Y is irreducible, then so is X and the map

$$\phi_\star : k(Y) \hookrightarrow k(X)$$

is an isomorphism.

In particular if X is irreducible and $U \subset X$ is a non-empty open subset, then there is a canonical isomorphism

$$k(X) \xrightarrow{\sim} k(U)$$

Proof. By (4.3.14) X is irreducible, so it is a special case of (4.3.15). ∎

Proposition 4.3.17 (Stalk Maps under Open Immersions I)
Let X be a local space with functions, $U \subset X$ an open subset and $W \subset X$ a irreducible closed subset such that $U \cap W \neq \emptyset$. Then $U \cap W$ is an irreducible subset of U and we have canonical isomorphisms of stalks making the following diagram commute

$$\begin{array}{ccc} \mathcal{O}_X(V) & \longrightarrow & \mathcal{O}_U(U \cap V) \\ \downarrow & & \downarrow \\ \mathcal{O}_{X,W} & \xrightarrow{\sim} & \mathcal{O}_{U,W\cap U} \end{array}$$

for all open $V \subset X$ for which $V \cap W \neq \emptyset$.

Proof. The inclusion $i : U \hookrightarrow X$ is an open immersion, so $U \cap W$ is irreducible by (4.1.73). This is then a special case of (4.3.15). ∎

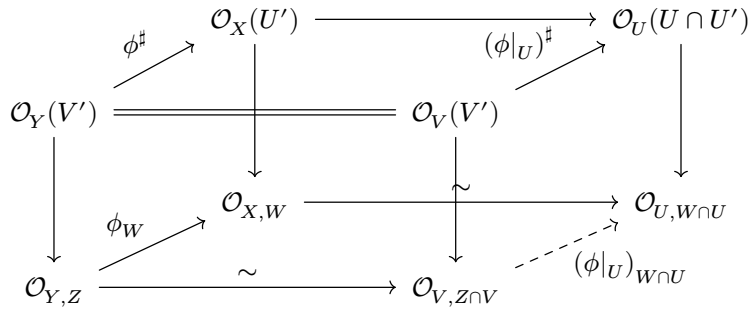
Lemma 4.3.18
Let U, Z be subsets of a topological space X, with U open. Then

$$\overline{Z} \cap U \subset \overline{Z \cap U}$$

Proof. Suppose $x \in \overline{Z} \cap U$ and V is an open subset containing x. Then $U \cap V$ is an open set containing x and therefore by (4.1.19) $Z \cap U \cap V \neq \emptyset$. As V was an arbitrary neighbourhood of x we conclude that $x \in \overline{Z \cap U}$. ∎

Proposition 4.3.19 (Restricting Stalk Maps to Open Subspaces)
Let $\phi : X \to Y$ be a regular morphism of local space with functions. Suppose $U \subset X$ and $V \subset Y$ are open subsets such that $U \subset \phi^{-1}(V)$, and $W \subset X$, $Z \subset Y$ are irreducible subsets such that $Z \subset \overline{\phi(W)}$ and $U \cap W \neq \emptyset$. Then $\phi(W \cap U) \subset \overline{Z \cap V}$ and there is a commutative cube for every open set $V' \subset V$ such that $V' \cap Z \neq \emptyset$ and $U' \subset \phi^{-1}(V')$ such that $U' \cap W \neq \emptyset$

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where the stalk isomorphisms are defined in (4.3.17) and $(\phi|_U)_{W\cap U}$ is the unique (local) homomorphism making the bottom square (or right side) commute

Proof. Observe by (4.3.18) that

$$\phi(W \cap U) \subset \phi(W) \cap V \subset \overline{Z} \cap V \subset \overline{Z \cap V}$$

which means the stalk map of $\phi|_U$ is well-defined. The sides of the cube are well-defined and commute by (4.3.17) and (4.3.5), and the top by definition. As the horizontal arrows are isomorphisms then we see that the bottom square commutes, and $(\phi|_U)_{W\cap U}$ is the unique homomorphism for which this is the case. ∎

4.3.3 Closed Immersions

Definition 4.3.20 (Closed Immersion)
*Let $\phi : X \to Y$ be a regular morphism of spaces with functions. We say that ϕ is a **closed immersion** if*

a) ϕ is a homeomorphism onto a closed subset $\phi(X)$ of Y
b) For all $x \in X$ we have the induced map on stalks

$$\mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_X$$

is surjective.

Proposition 4.3.21 (Closed Subset Structure Sheaf)
Let $(X, \mathcal{O}_X)$ be a space of functions and $Z \subset X$ a closed subset. Then there exists a unique sheaf of functions $\mathcal{O}_Z$ such that the inclusion $j : Z \to X$ is a closed immersion.

More precisely for $U \subset Z$ an open subset

$$\mathcal{O}_Z(U) := \left\{ f : U \to \overline{k} \;\middle|\; f(y) = g_i(y)\ \forall y \in Z \cap U_i \text{ for some } g_i \in \mathcal{O}_X(U_i) \text{ where } Z \cap \bigcup_{i\in I} U_i = U \right\}$$

Proof. Evidently j is a homeomorphism onto Z. We first show that j is regular. Suppose $g \in \mathcal{O}_X(U')$ then by definition we have $g \circ j \in \mathcal{O}_Z(j^{-1}(U')) = \mathcal{O}_Z(U' \cap Z)$.

Given $(V, f) \in \mathcal{O}_{Z,z}$ by taking a smaller open subset we may suppose that $f(y) = g(y)$ for all $y \in V$ for some $g \in \mathcal{O}_X(U)$ and $U \cap Z = V$. Then the image of (U, g) under the stalk map is $(U \cap Z, g \circ j)$ which is precisely (V, f). Therefore the stalk map j_z is surjective for all $z \in Z$.

Let $\mathcal{F}$ be a sheaf of functions for which j is a (regular) closed immersion and suppose $f \in \mathcal{O}_Z(U)$. Then $f|_{U_i\cap Z}(y) = g_i(y)$ for all $y \in Z \cap U_i$ and $g_i \in \mathcal{O}_X(U_i)$. By assumption $f|_{U_i\cap Z} = g_i \circ j \in \mathcal{F}(U_i \cap Z)$. By the sheaf condition we have $f \in \mathcal{F}(U)$ which shows $\mathcal{O}_Z$ is a subsheaf of $\mathcal{F}$.

We need to show that the inclusion of sheaves $\mathcal{O}_Z \hookrightarrow \mathcal{F}$ is an isomorphism. By (4.2.28) c) $\implies$ d) and (4.2.11) it is enough to show that it is surjective on stalks. However this follows by assumption by observing the composition $\mathcal{O}_{X,j(z)} \to \mathcal{O}_{Z,z} \to \mathcal{F}_z$ is surjective. ∎

Remark 4.3.22
This can be defined formally as $j^{-1}(\mathcal{O}_X/\mathcal{I})$ where $\mathcal{I}$ is the presheaf of regular functions which vanish on Z. We show in the Algebraic Geometry chapter's treatment of closed subvarieties that this is consistent with the concept of affine subvariety.

Proposition 4.3.23
Let $\phi : X \to Y$ be a regular morphism of spaces with functions. Then the following are equivalent

a) ϕ is a closed immersion

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b) $\phi(X)$ is closed and ϕ determines an isomorphism $X \xrightarrow{\sim} \phi(X)$ of spaces with functions using the structure defined in (4.3.21)

Proof. *a)* $\implies$ *b)* By definition $\phi(X)$ is closed and ϕ determines a homeomorphism onto $\phi(X)$. Let $Z \coloneqq \phi(X)$ and $\mathcal{O}_Z$ denote the canonical structure sheaf. There is a well-defined map

$$\mathcal{O}_Z(U) \to \mathcal{O}_X(\phi^{-1}(U))$$

which is injective, and therefore injective on stalks (4.2.28) c) $\implies$ d). By (4.2.11) we need only show that the map $\mathcal{O}_{Z,\phi(x)} \to \mathcal{O}_{X,x}$ is surjective. However by hypothesis the composite map

$$\mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_{Z,\phi(x)} \to \mathcal{O}_x$$

is surjective from which the result follows.

b) $\implies$ *a)* Evidently ϕ is a homeomorphism onto $\phi(X)$. Further we have a sequence of stalk maps

$$\mathcal{O}_{Y,\phi(x)} \xrightarrow{\sim} \mathcal{O}_{\phi(X),\phi(x)} \to \mathcal{O}_{X,x}$$

where the first is surjective by (4.3.21) and the second is an isomorphism by assumption. Therefore the composite is surjective and ϕ is a closed immersion. ∎

4.3.4 Locally Closed Subspace

Lemma 4.3.24 (Locally Closed Structure Sheaf)
Let $(X, \mathcal{O}_X)$ be a space with functions, $(Z, \mathcal{O}_Z)$ a closed subset with canonical structure defined in (4.3.21) and $(U, \mathcal{O}_U)$ an open subset with canonical structure (4.3.12). Then

$$\mathcal{O}_Z|_{Z\cap U} = \mathcal{O}_U|_{Z\cap U}$$

4.3.5 Glueing

Proposition 4.3.25 (Patching Topologies)
Let $X = \bigcup_{i=1}^n U_i$ be a set with topologies given for each U_i. Suppose that for all $i, j = 1...n$ the following properties hold

- *$U_i \cap U_j$ is open in U_i, and*
- *the identity map $1 : U_i \cap U_j \to U_i \cap U_j$ is continuous with respect to the subspace topologies on U_i and U_j respectively.*

Then there is a unique topology on X such that

- *the U_i are open in X,*
- *the topology on U_i coincides with the subspace topology from X*

More precisely it is given by

$$\{U \subset X \mid U \cap U_i \text{ open in } U_i \text{ for all } i = 1...n\}$$

Proposition 4.3.26 (Glueing Topologies)
Let $X = \bigcup_{i=1}^n U_i$ be a set together with bijections $\theta_i : U_i \xrightarrow{\sim} W_i$ where W_i are topological spaces. Suppose that for all $i, j = 1...n$ the following properties hold

- *$\theta_i(U_i \cap U_j)$ is open in W_i,*
- *the map $\theta_{ij} \coloneqq \theta_j \circ \theta_i^{-1} : \theta_i(U_i \cap U_j) \to \theta_j(U_i \cap U_j)$ is continuous.*

Then there is a unique topology on X such that

- *U_i are open in X,*
- *θ_i is a homeomorphism with respect to the subspace topology on U_i for all $i = 1...n$.*

More precisely the topology is given by

$$\{U \subset X \mid \theta_i(U \cap U_i) \text{ is open in } W_i \text{ for all } i = 1...n\}$$

Proposition 4.3.27 (Glueing Regular Maps)
Let X, Y be spaces with functions and $X = \bigcup_{i=1}^n U_i$ an open cover. Suppose there exists regular maps

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$$\phi_i : \left(U_i, \mathcal{O}_X|_{U_i}\right) \to (Y, \mathcal{O}_Y)$$

which satisfy $\phi_i|_{U_i \cap U_j} = \phi_j|_{U_i \cap U_j}$ *for all* $i, j = 1...n$*. Then there exists a unique map* $\phi : X \to Y$ *such that* $\phi|_{U_i} = \phi_i$*, which is regular.*

Proof. As a function ϕ is clearly well-defined and unique. It is continuous by (4.1.29). Suppose $g \in \mathcal{O}_Y(V)$ then by definition $(g \circ \phi)|_{\phi^{-1}(V) \cap U_i} = g \circ \phi_i \in \mathcal{O}_{U_i}(\phi^{-1}(V) \cap U_i) = \mathcal{O}_X(\phi^{-1}(V) \cap U_i)$. As $\mathcal{O}_X$ is a sheaf this shows that $g \circ \phi$ is regular as required. ∎

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